

GENOWL STUDIO

System Blueprint, Product Insights & The 3D WebGL Production Roadmap

Executive Lead: Antriksh Soun (Genowl)	Project Version: Genowl 2.0 (Production)
Official Channel: genowlai@gmail.com	Production Staging: Hostinger Cloud
Architecture: React 18 + Express + Supabase + WebGL	Target Turnaround: 48 - 72 Hours (100% IP Transfer)

1. Executive Vision & Core Philosophy

Our Foundational Principle:
"In today's world, everybody knows that for almost every service possible there is an AI tool. But of course people don't have the time to use and master every tool. That is exactly why you choose Genowl: all you have to do is buy our service and tell us what to build — the rest is on us."

Transparent Service Structure & Pricing Tiers

Tier / Service	Price	Turnaround	Core Deliverables & Specifications
2D Responsive Website	\$500 Flat	48 - 72h	High-converting, mobile-first responsive architecture, semantic React/Vite code, SEO 95+, 100% full IP ownership.
3D Interactive WebGL	\$2,500 Flat	48 - 72h	Cinema-grade 3D digital worlds, Three.js canvas, scroll-driven camera choreography, GLTF Draco compression, 60 FPS mobile/desktop GPU optimization.
Video Generation Sprint	\$99 Flat	24 - 48h	High-impact commercials, promotional reels, and brand visual narratives rendered via cutting-edge diffusion engines.
Personalized AI Automation	\$99 Flat	24 - 48h	Custom customer service chatbots, intelligent context engineering, operational workflow automation.
Content Creation Package	\$99 Flat	24 - 48h	Conversion sales copywriting, branded social media asset suites, editorial calendars, launch articles.

2. Technical Architecture & Engineering Decisions

Over our development cycle, every component was meticulously engineered to provide enterprise security, flawless cross-device ergonomics, and frictionless customer conversions:

A. Fortified Master Admin Security:

- **Permanently Locked Master Password:** Standardized across the entire platform to CristianoMessi@2005.
- **Zero Setup Vulnerability:** The initial 'Setup Master Password' UI was permanently deleted. Any session on any domain, mobile phone, or desktop immediately requires master credentials.
- **Web Modification Disabled:** Password updating forms were removed from the client interface to guarantee no unauthorized browser overwrites.

- **Universal Access:** Accessible directly from both Desktop Navbar, Mobile Drawer, and persistent Footers.

B. Mandatory Account Authentication for Inquiries & Orders:

- **Spam & Impersonation Prevention:** Visitors must be logged into a verified account before submitting project briefs or problem tickets.
- **Identity Binding:** Submissions automatically lock to the client's verified name and email, presenting personalized confirmations (*"Thank you, [Name]!"*).

C. Validated Phone / WhatsApp Capture:

- **Glitch-Free Input:** Engineered to accept international numbers (e.g. +91 98765 43210) without cursor jumping or formatting bugs.
- **Direct Lead Stream:** Validated phone numbers are sent directly to the database and email forwarding desk for immediate WhatsApp/phone outreach.

D. Central Cloud Database (Supabase PostgreSQL):

- **Global Real-Time Sync:** Orders and inquiries stream live to Supabase tables (genowl_orders, genowl_inquiries, genowl_users).
- **Permanent Lead Vault:** Client inquiries and phone numbers are safely stored in cloud PostgreSQL, protected from browser cache wipes.

E. Bulletproof Lead Forwarding to Official Gmail:

- **Dual-Channel Architecture:** Employs zero-config direct webhook dispatch via FormSubmit directly to genowlai@gmail.com combined with Resend API fallback.
- **Branded Client Confirmation:** Clients receive automated HTML receipts embedded with the official Golden Owl insignia.

3. The Proprietary 3D WebGL Production Roadmap ("Our Way")

At Genowl Studio, our **\$2,500 3D Website tier** is engineered to position brands in the top 1% of the internet. Rather than relying on generic page builders or heavy, laggy assets, we employ a battle-tested, cinema-grade WebGL architecture. Here is our complete step-by-step roadmap for building 3D interactive experiences:

Phase 1: Creative Concept & Camera Storyboarding

- **Scroll Narrative Definition:** Mapping the brand's core pitch to an interactive journey.
- **Virtual Spline Design:** Defining 3D camera translation (X, Y, Z) and rotation vectors parameterized between scroll values 0.0 (entry) to 1.0 (conclusion).
- **Focal Keyframes:** Aligning 3D object angles with high-converting copy and CTA buttons.

Phase 2: 3D Asset Modeling & Texture Baking Pipeline

- **Precision Hard-Surface & Organic Modeling:** Sculpted in Blender / Cinema4D with strict polygon budgets (< 50,000 tris).
- **PBR Texture Baking:** Consolidating Diffuse, Normal, Roughness, Metallic, and Ambient Occlusion into unified 2K atlases.
- **Draco & Meshopt Compression:** Compressing raw 80MB 3D meshes down to 1.5MB - 3.5MB GLTF/GLB containers for instantaneous web loading.

Phase 3: Three.js & WebGL Canvas Engine

- **Non-Blocking Background Rendering:** Isolated Three.js WebGLRenderer mounted on a fixed, pointer-transparent canvas.
- **Custom GLSL Shaders:** Atmospheric fog, particle dust dynamics, refractive glass, and glowing energy fields.
- **Cinematic Post-Processing:** Tone mapping (ACESFilmic), selective bloom on emissive elements, and subtle depth-of-field blur.

Phase 4: Hybrid Canvas + DOM Layering (The Secret Sauce)

- **Zero-Jank Lerp Physics:** Smooth linear interpolation (lerp factor: 0.05 - 0.08) tying mouse wheel and touch drags to camera translation.
- **Decoupled HTML Overlay:** All headings, prices (\$500 / \$2,500), and forms live in semantic React DOM layers above the canvas, ensuring 100% SEO indexing and accessibility.
- **Scroll-Triggered Section Gates:** Kinetic text reveals synchronize precisely as 3D models orbit into focal position.

Phase 5: Mobile GPU Ergonomics & 60 FPS Performance Budget

- **DPR Clamping:** Clamping device pixel ratio to $\text{Math.min}(\text{window.devicePixelRatio}, 2)$ to prevent thermal throttling on 4K mobile displays.
- **Dynamic LOD & Shader Downsampling:** Automatically simplifying complex geometry and disabling heavy post-processing on mobile GPUs.
- **Touch Gesture Normalization:** Fluid vertical touch scrolling with `-webkit-overflow-scrolling: touch` and zero horizontal viewport drift.

Phase 6: Cloud Deployment & Client Handover

- **CDN Edge Caching:** GLB assets and frame sequences cached globally via HTTP stale-while-revalidate headers.
- **100% Full IP Transfer:** Client receives full React/Vite source code, Blender 3D master files, textures, and deployment documentation within 48 to 72 hours.

4. System Specification Summary

Component Layer	Technology / Implementation	Function & Benefit
Frontend Framework	React 18 + Vite + TypeScript	Instant HMR, typed component boundaries, sub-4s production builds.
Styling Engine	Tailwind CSS + Custom Obsidian Tokens	Bespoke glassmorphism, responsive touch ergonomics, zero runtime overhead.
Interactive 3D Engine	Canvas 240-Frame Lerp + Three.js WebGL	Cinema-grade visual fidelity running at 60 FPS across mobile and desktop.
Cloud PostgreSQL	Supabase Cloud Database (RLS Secured)	Real-time persistent lead storage for inquiries, phone numbers, and orders.
Lead Dispatch Desk	FormSubmit Webhook + Resend REST API	Guaranteed inbox arrival to genowlai@gmail.com with zero configuration.
Payment Processing	Razorpay Indian & International Gateway	Instant UPI (GPay, PhonePe), RuPay/Visa/MasterCard, and Net Banking.
Hosting & Deployment	Hostinger Cloud + Express Server Bridge	Automatic Git CI/CD deployments directly from GitHub main repository.

Genowl Studio Engineering Commitment:

This document establishes the official architectural baseline and operational standard for Genowl Studio. Every client website, 3D experience, and AI integration produced follows these guidelines strictly to ensure unmatched speed, cinema-grade aesthetics, and sustainable client growth.